

3.2.3 Section C: The challenge of resource management

In this section, students are required to study [Resource management](#) (page 22) and **one** from [Food](#) (page 23) or [Water](#) (page 24) or [Energy](#) (page 25).

3.2.3.1 Resource management

Key idea	Specification content
Food, water and energy are fundamental to human development.	The significance of food, water and energy to economic and social well-being. An overview of global inequalities in the supply and consumption of resources.

Key idea	Specification content
The changing demand and provision of resources in the UK create opportunities and challenges.	An overview of resources in relation to the UK. Food: - the growing demand for high-value food exports from low income countries and all-year demand for seasonal food and organic produce - larger carbon footprints due to the increasing number of 'food miles' travelled, and moves towards local sourcing of food - the trend towards agribusiness. Water: - the changing demand for water - water quality and pollution management - matching supply and demand – areas of deficit and surplus - the need for transfer to maintain supplies. Energy: - the changing energy mix – reliance on fossil fuels, growing significance of renewables - reduced domestic supplies of coal, gas and oil - economic and environmental issues associated with exploitation of energy sources.

3.2.3.2 Food

Key idea	Specification content
Demand for food resources is rising globally but supply can be insecure, which may lead to conflict.	Areas of surplus (security) and deficit (insecurity): - global patterns of calorie intake and food supply - reasons for increasing food consumption: economic development, rising population - factors affecting food supply: climate, technology, pests and disease, water stress, conflict, poverty. Impacts of food insecurity – famine, undernutrition, soil erosion, rising prices, social unrest.

Key idea	Specification content
Different strategies can be used to increase food supply.	Overview of strategies to increase food supply: • irrigation, aeroponics and hydroponics, the new green revolution and use of biotechnology, appropriate technology • an example of a large scale agricultural development to show how it has both advantages and disadvantages. Moving towards a sustainable resource future: • the potential for sustainable food supplies: organic farming, permaculture, urban farming initiatives, fish and meat from sustainable sources, seasonal food consumption, reduced waste and losses • an example of a local scheme in an LIC or NEE to increase sustainable supplies of food.

3.2.3.3 Water

Key idea	Specification content
Demand for water resources is rising globally but supply can be insecure, which may lead to conflict.	Areas of surplus (security) and deficit (insecurity): • global patterns of water surplus and deficit • reasons for increasing water consumption: economic development, rising population • factors affecting water availability: climate, geology, pollution of supply, over-abstraction, limited infrastructure, poverty. Impacts of water insecurity – waterborne disease and water pollution, food production, industrial output, potential for conflict where demand exceeds supply.

Key idea	Specification content
Different strategies can be used to increase water supply.	Overview of strategies to increase water supply: - diverting supplies and increasing storage, dams and reservoirs, water transfers and desalination - an example of a large scale water transfer scheme to show how its development has both advantages and disadvantages. Moving towards a sustainable resource future: - water conservation, groundwater management, recycling, 'grey' water - an example of a local scheme in an LIC or NEE to increase sustainable supplies of water.

3.2.3.4 Energy

Key idea	Specification content
Demand for energy resources is rising globally but supply can be insecure, which may lead to conflict.	Areas of surplus (security) and deficit (insecurity): - global distribution of energy consumption and supply - reasons for increasing energy consumption: economic development, rising population, technology - factors affecting energy supply: physical factors, cost of exploitation and production, technology and political factors. Impacts of energy insecurity – exploration of difficult and environmentally sensitive areas, economic and environmental costs, food production, industrial output, potential for conflict where demand exceeds supply.

Key idea	Specification content
Different strategies can be used to increase energy supply.	Overview of strategies to increase energy supply: • renewable (biomass, wind, hydro, tidal, geothermal, wave and solar) and non-renewable (fossil fuels and nuclear power) sources of energy • an example to show how the extraction of a fossil fuel has both advantages and disadvantages. Moving towards a sustainable resource future: • individual energy use and carbon footprints. Energy conservation: designing homes, workplaces and transport for sustainability, demand reduction, use of technology to increase efficiency in the use of fossil fuels • an example of a local renewable energy scheme in an LIC or NEE to provide sustainable supplies of energy.